

Prime-factor densities: bounded-gap spectra and a Riemann hypothesis criterion

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Abstract

We study reversals in the density of the k th smallest distinct prime factor and the finer arithmetic error in its consecutive-coefficient ratio. For fixed local laws with $\eta_p^{-1} = p/\nu + \beta + o(1)$, the finite maximal separated reversal count has exact rate $\limsup_k k^{-1} \log \log(3 + N_\eta(k)) = \sup_{H \geq 2} \gamma_H / (H + \nu)$, where γ_H records bounded-gap frequency. This spectrum has a finite phase profile and is realized by products of distinct affine forms in the sum of two primes. For the ordinary law, a uniform comparison with a full entire-function reference transfers the classical Nicolas prime-product error to the actual coefficient ratio. For each fixed positive moving-rank parameter, eventual negativity of the centered ratio is equivalent to the Riemann hypothesis. We give quantitative excursions under false RH and sharp reference-location and reversal-count bounds under RH. A separate Hardy–Littlewood-shaped twin-count hypothesis places the last actual ascent at relative distance tending to zero from its potential cutoff. We distinguish classical analytic inputs from the potentially new spectrum and coefficient transfer. The main maintained results are formalized in Lean; no new prime-gap bound or proof of RH is claimed.

1 Introduction and historical context

Let $d_k(p)$ be the natural density of positive integers whose k th smallest *distinct* prime factor is p . The question whether $p \mapsto d_k(p)$ first rises and then falls is now called Erdős Problem 690. Our subject is the finer structure left after unimodality fails: the number of separated reversals, their possible locations, and the arithmetic information encoded by their growth.

Erdős formulated the question in 1979 [1, p. 75]. The same paper discusses the typical scale $\log \log p_k(n) \sim k$ for prime factors of integers. This

concerns where probability mass lies, not the mode of the individual-prime density. Erdős–Tenenbaum [2, Corollary 5] later corrected the earlier assertion about the mode: a maximizing prime satisfies $\log p \sim k/\log k$, rather than $\log p \sim k$. Neither statement counts reversals. The occurrence of a double exponential in classical prime-factor distribution does not by itself supply the theorem below.

Cambie [3, Theorem 5] proved unimodality for $k = 1, 2, 3$ and verified its failure for $4 \leq k \leq 20$. His Theorem 3 gives superpolynomially many local maxima for the different density $\delta_1(n, m)$ in Erdős Problem 692: the density of integers having exactly one divisor in $\{n + 1, \dots, m - 1\}$. Its moving variable is m , with n fixed. It is not a reversal count for $d_k(p)$.

Wang–Crapis [4, Theorem 1.1] proved non-unimodality for every $k \geq 4$. Their proof compares a symmetric-polynomial ratio with prime gaps and produces a descent followed by an ascent. We extend the comparison to local laws and make the construction quantitative. The question here is how the maximal number of witnesses grows with k . Its exact rate also requires a converse: a sufficiently late ascent must come from an appropriately short gap.

There are two main contributions. The first is the exact reversal frequency spectrum, its arithmetic realization, and its inverse late-ascent interpretation. Quantitative bounded-gap results are existing analytic inputs, not new sieve results. The second is a uniform transfer of a classical RH-sensitive error into the actual moving-rank coefficient ratio, preserving its sign at a scale much smaller than any fixed inverse power of $\log \log x$.

Nicolas’s primorial criterion and its signed error [6, 7] supply the analytic source. The finite generating product and its saddle-point analysis already occur in Erdős–Tenenbaum [2]; the normalized Euler product also appears in mod-Poisson theory [8, 9]. We claim neither the generating function nor the Nicolas spectral expansion as new. The potentially new statements are the exact gap-frequency spectrum and the quantitative comparison with a full theta-centered coefficient reference, together with their stated applications. Priority remains provisional pending independent review.

Sections 4–8 prove the unconditional spectrum. Section 9 treats recurrence and the separate Hardy–Littlewood conditional location result. Sections 10–12 establish the coefficient transfer, RH criteria and reference capacities. The public repository contains maintained proofs and a provenance ledger. This is an AI-assisted research draft; independent expert review and assessment of novelty remain outstanding.

2 The argument in plain English

Choose a position in an integer's ordered list of prime factors, say the tenth distinct prime factor. For each prime, ask what fraction of sampled integers put that prime in the chosen position. These fractions form a sequence. A reversal means that the sequence drops and later rises again. We count occurrences separated far enough that they do not reuse the same stretch of the sequence.

Two effects compete. Moving to a larger prime lowers its own chance of dividing the integer. At the same time, the additional smaller primes make it easier to have the required number of preceding factors. An exact formula measures this competition. On the relevant scale, a short gap between consecutive primes favors a rise, whereas a long gap favors a drop.

The useful measuring scale is the logarithm of the logarithm of the prime. On that scale the available rise threshold is approximately the rank divided by the scale. This explains the double-exponential locations in the argument. Eventually even the shortest possible prime gap is too large for a rise, so every fixed rank still has a final decreasing tail.

Many short gaps alone are not enough to count reversals. We must put a long gap before each selected short gap and keep the witnesses distinct. Periodic blocks of composite integers provide barriers that consecutive primes must jump. Grouping short gaps between the barriers and taking alternate occupied groups produces separated witnesses, losing only a fixed factor in their number.

Finally, different short-gap sizes can have different frequencies. Each size contributes its frequency divided by a threshold cost. The greatest contribution is exactly the reversal growth rate. For products of affine prime sums, the number of distinct factors changes that cost, while finitely many exceptional primes change only the initial details.

A second, much finer comparison uses the entire prime-product profile as a reference. The ratio that decides an ascent differs from this reference by a small signed error. We show that this error carries the same RH information as Nicolas's primorial inequality. Under RH it is eventually negative, with a bias from prime squares. If RH is false it has larger excursions of both signs. The rank moves outward with the prime cutoff in this statement; it does not contradict eventual decline at each fixed rank.

This determines how the prefix changes the threshold for a possible ascent. An actual ascent also needs an actual prime gap below that threshold. RH controls the fine reference error, while a Hardy–Littlewood twin-count assumption supplies twins near the potential endpoint. Their roles and their

different precision are kept separate throughout.

3 Definitions and main results

All logarithms are natural. Index the primes by $p_0 = 2, p_1 = 3, \dots$, and write $\mathcal{L}(x) = \log \log x$ for $x > 1$ and $\mathcal{E}(t) = \exp(\exp t)$. Ranks k are positive integers. A fixed *local law* consists of probabilities $0 \leq \eta_p \leq 1$. Define the finite products

$$M_r(p) = [z^r] \prod_{q < p} (1 - \eta_q + \eta_q z), \quad D_{\eta,k}(p) = \eta_p M_{k-1}(p), \quad (1)$$

with $M_r = 0$ for $r < 0$. For the ordinary law $\eta_p = 1/p$, the Chinese remainder theorem identifies $D_{\eta,k}$ with d_k . The second canonical law is

$$\eta_2^{\text{odd}} = 0, \quad \eta_p^{\text{odd}} = \frac{p-2}{(p-1)^2} \quad (p > 2). \quad (2)$$

Definition 3.1. A reversal at rank k consists of indices $d < a$ with

$$D_{\eta,k}(p_{d+1}) < D_{\eta,k}(p_d), \quad D_{\eta,k}(p_a) < D_{\eta,k}(p_{a+1}).$$

A family of m such witnesses is separated if $a_i + 1 < d_j$ for $i < j$. Its maximal attainable size, when finite, is $N_\eta(k)$.

This counts raw-density descent-ascent witnesses, not oscillations of a centered error or a smoothed approximation.

For integers $H \geq 0$, set

$$\begin{aligned} G_H(X) &= \#\{i : p_i \leq X, p_{i+1} - p_i \leq H\}, \\ \gamma_H &= \limsup_{\substack{X \rightarrow \infty \\ X \in \mathbb{N}}} \frac{\log \log(3 + G_H(X))}{\mathcal{L}(X)}, \\ \Lambda(\nu) &= \sup_{H \geq 2} \frac{\gamma_H}{H + \nu} \quad (\nu > 0). \end{aligned} \quad (3)$$

The 3 keeps the logarithms defined when counts vanish. We have $0 \leq \gamma_H \leq 1$ and $\gamma_0 = \gamma_1 = 0$.

Theorem 3.2 (Exact reversal spectrum). *Suppose $\nu > 0$, $\beta \in \mathbb{R}$, and, along sufficiently large primes,*

$$0 < \eta_p < 1, \quad \eta_p^{-1} = p/\nu + \beta + \rho(p), \quad \rho(p) \longrightarrow 0. \quad (4)$$

Every fixed-rank sequence $D_{\eta,k}$ is eventually nonincreasing, its reversal maximum is finite and attained, and

$$\limsup_{k \rightarrow \infty} \frac{\log \log(3 + N_{\eta}(k))}{k} = \Lambda(\nu) > 0. \quad (5)$$

The law, its parameters and its exceptional coordinates are fixed before k tends to infinity.

The weaker estimate $\eta_p = \nu/p + O(p^{-2})$ alone does not supply the vanishing difference of reciprocal errors used in this theorem.

Theorem 3.3 (Canonical bounds). *For either canonical law and every $\varepsilon > 0$, for all sufficiently large k ,*

$$\mathcal{E}(k/602) \leq N_{\eta}(k) \leq \mathcal{E}((1/3 + \varepsilon)k).$$

Each fixed-rank canonical sequence is eventually strictly decreasing.

The positive constant uses a conservative quantitative width-600 input; it is not a claim about the best known numerical prime-gap bound.

Theorem 3.4 (Affine arithmetic realization). *Fix $m \geq 1$, positive integers c_j , and integers d_j , with $c_i d_j \neq c_j d_i$ for $i \neq j$. Put*

$$F(p, q) = \prod_{j=1}^m (c_j(p + q) + d_j).$$

Sample all ordered prime pairs $p, q \leq X$, including the diagonal, with denominator $\pi(X)^2$. At every fixed rank k and prime ℓ , the probability that ℓ is the k th smallest distinct prime factor of $|F(p, q)|$ has a limit $D_{F,k}(\ell)$; zero outputs are excluded from the event. The maximal separated reversal count $N_F(k)$ is finite and attained, and

$$\limsup_{k \rightarrow \infty} \frac{\log \log(3 + N_F(k))}{k} = \Lambda(m) > 0.$$

The size limit comes first, with family, rank and prime fixed. Outputs are weighted by their prime-pair representations.

The second main result, Theorem 11.1, concerns the ordinary coefficient ratio at rank $r = \lfloor \alpha(\gamma + \log \log \theta(x)) \rfloor$, for each fixed $\alpha > 0$. Its precise full-profile normalization is defined in Section 10. Eventual negativity of that centered ratio is equivalent to RH. This statement is separate from Theorem 3.2, whose hypotheses contain no RH assumption.

4 The exact comparison and uniform estimates

4.1 First differences

For consecutive primes $p < q$ with interior probabilities,

$$M_r(q) = (1 - \eta_p)M_r(p) + \eta_p M_{r-1}(p).$$

Direct subtraction gives

$$D_{\eta,k}(q) - D_{\eta,k}(p) = \eta_p \eta_q \{M_{k-2}(p) - T_\eta(p, q)M_{k-1}(p)\}, \quad (6)$$

where $T_\eta(p, q) = \eta_q^{-1} - \eta_p^{-1} + 1$. When $M_{k-1}(p) > 0$, an ascent is equivalent to $M_{k-2}(p)/M_{k-1}(p) > T_\eta(p, q)$. For the ordinary law, writing $R = M_{k-2}/M_{k-1}$ and $g = q - p$, the exact real-valued gap threshold is $R - 1$: a strict ascent occurs if and only if $g < R - 1$. Equality gives a flat step. This threshold does not constrain the actual gap to be small. The ordinary comparison level is $q - p + 1$; for the odd law at $p > 2$ it is

$$q - p + 1 - \frac{q - p}{(p - 2)(q - 2)}.$$

4.2 Finite exceptions and symmetric polynomials

Only finitely many coordinates in (4) can be forced ($\eta_p = 1$) or excluded ($\eta_p = 0$). Let f count the forced coordinates. Past them, each contributes a factor z to the generating polynomial; excluded coordinates contribute 1. Retain all interior coordinates, including the exceptional ones, and set

$$w_p = \frac{\eta_p}{1 - \eta_p}, \quad S(p) = \sum_{\substack{q < p \\ 0 < \eta_q < 1}} w_q.$$

For $k > f$, put $r = k - 1 - f$. The common positive product of $1 - \eta_q$ cancels in the ratio, leaving $E_{r-1}(p)/E_r(p)$, where E_r is the elementary symmetric polynomial of degree r in the finite list of interior odds.

Lemma 4.1. *For at least $r \geq 1$ positive weights, let W_{r-1} be the largest sum of $r - 1$ distinct weights. Then*

$$\frac{r}{S} \leq \frac{E_{r-1}}{E_r} \leq \frac{r}{S - W_{r-1}}, \quad (7)$$

and the right denominator is positive.

Proof. The terms of SE_{r-1} which choose a new index total rE_r . The remaining terms are

$$\sum_{|A|=r-1} \left(\sum_{j \in A} w_j \right) \prod_{j \in A} w_j,$$

between zero and $W_{r-1}E_{r-1}$. Rearrangement proves the bounds. At least one positive weight remains after any $r-1$ are removed. $\square$

This specializes to Wang–Crapis [4, Lemmas 3.1–3.2]; the present use also retains the exact forced-rank shift.

Expanding (4) gives $w_p = \nu/p + O(p^{-2})$. The reciprocal-prime estimate and the convergent quadratic tail give

$$S(p) = \nu\mathcal{L}(p-1) + O_\eta(1), \quad W_{r-1} = O_\eta(1 + \log(r+1)). \quad (8)$$

For the second estimate, dominate all but finitely many odds by C/p and use $p_j \geq j+2$. The finitely many exceptional odds contribute a fixed constant; their order need not be monotone.

Lemma 4.2 (Uniform sign margins). *Fix $0 < u \leq v$. Uniformly in the band $uk \leq \mathcal{L}(p-1) \leq vk$, as $k \rightarrow \infty$,*

$$\frac{E_{r-1}(p)}{E_r(p)} = \frac{k}{\nu\mathcal{L}(p-1)} + o(1), \quad r = k-1-f.$$

Gaps at most H give strict ascents throughout the band if $(H+\nu)v < 1$. Gaps at least D give strict descents there if $Du > 2$, for sufficiently large rank and lower prime.

Proof. The prime number theorem ensures more than r interior primes below p uniformly in this band. Equations (7) and (8) have numerator $k + O_\eta(1)$ and denominator $\nu\mathcal{L}(p-1) + O_\eta(\log(k+1))$, bounded below by a positive multiple of k . This proves the uniform ratio statement.

Writing $g = q - p$, the threshold is $g/\nu + 1 + \rho(q) - \rho(p)$. Its error is bounded by twice the supremum of $|\rho|$ beyond p , even for large g . For $g \leq H$ the limiting lower ratio is $1/(\nu v) > (H+\nu)/\nu$. For $g \geq D$ the upper ratio is $1/(\nu u) + o(1) < D/\nu$. Both margins are strict. $\square$

Without an upper band endpoint, the same estimates give a one-sided bound: on $\mathcal{L}(p-1) \geq bk$ the ratio is at most $1/(\nu b) + o(1)$, uniformly for fixed $b > 0$.

4.3 Every fixed rank

For fixed $r \geq 1$, the ratio tends to zero by (7)–(8). Consecutive sufficiently large primes have gap at least two, so their threshold is uniformly positive. Equation (6) gives strict descent for supported degrees. If $k \leq f$, the tail is zero. If $k = f + 1$, the remaining degree is zero, its predecessor coefficient vanishes, and the same identity again gives descent. Thus every fixed-rank tail is nonincreasing. For the ordinary law, eventual decrease was already noted by Cambie [3, §3]. The extension here retains the forced and excluded coordinates.

Once all ascent indices lie below an integer I , a separated family has size at most I . Attainable sizes form a nonempty bounded subset of $\mathbb{N}$ (zero is attainable), so their maximum is attained. The canonical laws have no forced coordinates and have positive supported tails, giving strict eventual decrease.

5 From short gaps to separated reversals

The following finite argument supplies the counting step and its loss. It turns actual small-gap supply into separated raw reversals. The later RH coefficient correction is not an input to this construction.

Lemma 5.1 (Periodic composite barriers). *Fix integers $D \geq H + 1$ and $M = D!$. Let $\mathcal{S}$ be a nonempty finite set of integer starts such that*

$$X \leq n, \quad n + H \leq Y, \quad \#([n, n + H] \cap \{\text{primes}\}) \geq 2, \quad X \geq M + 2.$$

There are m separated pairs of consecutive-prime gaps, a gap at least D followed by one at most H , all endpoints in $[X, Y]$, with

$$|\mathcal{S}| \leq 2(M + H)(m + 1). \tag{9}$$

Proof. Choose consecutive primes $p < q$ in each window, and assign them to $c = \lfloor (p - 2)/M \rfloor$. Since $cM + j$ is composite for $2 \leq j \leq D$, and $c \geq 1$,

$$cM + D + 1 \leq p < q \leq (c + 1)M + 1.$$

Indeed a prime q beyond the last endpoint but within $H < D$ of p would lie in the next composite block. A short pair cannot cross that barrier.

Starts assigned to cell c lie between $cM + D + 1 - H$ and $(c + 1)M + 1$, so there are at most $M + H$ of them. Order the occupied cells $c_0 < \dots < c_{t-1}$ and choose one pair in each, with lower-prime index u_j . Between chosen

primes in two successive occupied cells, a consecutive-prime gap crosses the destination cell's whole composite block. Its length is at least D , and it precedes the chosen short gap there.

Take destination cells $c_2, c_4, \dots, c_{2m}$, where $m = \lfloor (t-1)/2 \rfloor$. Choose each crossing after the lower prime selected in the preceding occupied cell. The unused occupied cell between successive destinations ensures $a_i + 1 < d_{i+1}$. Every crossing lies between selected endpoints, so remains in $[X, Y]$. Finally $t \leq 2(m+1)$ and $|\mathcal{S}| \leq (M+H)t$, proving (9). $\square$

No uniform distribution among cells is needed. Empty stretches are harmless, since a crossing still exists between selected occupied cells.

Fix $u, v > 0$ with $(H+\nu)v < 1$ and choose an integer $D \geq H+1$ with $D > 2/u$. If all primes from X through $Y+H$ lie in the sign band, the two lemmas imply, at sufficiently large k ,

$$G_H(Y) \leq X + B(N_\eta(k) + 1), \quad B = 2(D! + H). \quad (10)$$

At most X lower primes are deleted below X ; each retained short gap supplies a window starting at its lower prime. The upper endpoint is $Y+H$, since a counted gap may end above Y . All constants are fixed before k grows.

Corollary 5.2 (Positive-power transfer). *Suppose that for fixed H and $\delta > 0$, for every sufficiently large X at least X^δ integer starts in $[X, 2X]$ have two primes in their width- H windows. For every $0 < a < 1/(H+\nu)$,*

$$N_\eta(k) \geq \mathcal{E}(ak) \quad \text{for all sufficiently large } k.$$

Proof. Choose $a < b < 1/(H+\nu)$ and $X = \lceil \mathcal{E}(bk) \rceil$. Every endpoint in $[X, 2X+H]$ has $\mathcal{L}(p-1) = bk + o(1)$. Fixed $u < b < v < 1/(H+\nu)$ therefore provide the sign band. The cell lemma gives $N_\eta(k) + 1 \geq X^\delta/B$. Since $\log(X^\delta/B) = \delta \exp(bk) + O(1)$ and $a < b$, the right side eventually exceeds $1 + \mathcal{E}(ak)$. $\square$

We use the existing quantitative input

$$\#\{n \in [X, 2X] \cap \mathbb{N} : \#([n, n+600] \cap \{\text{primes}\}) \geq 2\} \geq X^{1/3} \quad (11)$$

for all sufficiently large X . Its maintained declaration is `PrimeGaps.eventually_many_prime_windows_600`. Section 14 records its provenance. Quantitative prime-cluster theory, including Maynard's work [5], is the analytic background. We use this count, not merely infinitude of bounded gaps. With $\nu = 1$ the corollary applies at $a = 1/602 < 1/601$.

6 Proof of the exact spectrum

The two directions preserve the frequency subsequences and each individual gap's cutoff. This matching rate, rather than the presence of a double exponential alone, is the principal unconditional result.

6.1 Lower bound: arbitrary frequency subsequences

Fix H and $0 < a < \gamma_H/(H + \nu)$. Choose

$$0 < g < \gamma_H, \quad 0 < v_0 < v < \frac{1}{H + \nu}, \quad a < gv_0,$$

then $0 < u < a$ and $a < b < gv_0$. There are arbitrarily large integers Y with $3 + G_H(Y) > \mathcal{E}(g\mathcal{L}(Y))$. Assign $k = \lceil \mathcal{L}(Y)/v_0 \rceil$. The bounds $v_0k - v_0 < \mathcal{L}(Y) \leq v_0k$ imply $3 + G_H(Y) > \mathcal{E}(bk)$ at arbitrarily large assigned ranks.

Take $X = \lceil \mathcal{E}(uk) \rceil + 1$. The primes between X and $Y + H$ lie in a band with positive lower coefficient and upper coefficient v . Equation (10) applies. If $N_\eta(k) < \mathcal{E}(ak)$, then

$$3 + G_H(Y) \leq 3 + X + B(1 + \mathcal{E}(ak)) < \mathcal{E}(bk)$$

eventually, a contradiction. Therefore $N_\eta(k) \geq \mathcal{E}(ak)$ for arbitrarily large k . Supremizing over H proves the lower bound $\Lambda(\nu)$ in (5). The argument keeps the actual frequency subsequence; it does not replace it by an all-large- X hypothesis.

6.2 Upper bound: each gap retains its own cutoff

For fixed integer $g \geq 2$ and $\varepsilon > 0$, the one-sided ratio bound implies that an ascent across a gap of size g must satisfy, apart from fixed initial exceptions,

$$\mathcal{L}(p - 1) < \left(\frac{1}{g + \nu} + \varepsilon \right) k. \quad (12)$$

The same estimate excludes all gaps greater than J above $\mathcal{L}(p - 1) = uk$ if $1/(J + 1 + \nu) < u$. In both statements, the reciprocal error is uniformly small beyond the lower prime.

Consequently

$$N_\eta(k) \leq T_k + \sum_{2 \leq g \leq J} G_g(U_g(k)), \quad (13)$$

where we may take

$$T_k = \lceil \mathcal{E}(uk) \rceil + O(1), \quad U_g(k) = \left\lceil \mathcal{E} \left(\left(\frac{1}{g+\nu} + \varepsilon \right) k \right) \right\rceil + 1.$$

Each witness has a distinct ascent index. There are at most T_k initial ones; all others have an actual gap $g \leq J$ and are counted in that gap's term. Overcounting by the cumulative G_g only strengthens the upper bound.

Let $A > \Lambda(\nu)$. Choose $0 < u < A$ and then J as above. For each fixed g and $\delta > 0$, eventually

$$\log \log(3 + G_g(U)) \leq (\gamma_g + \delta) \mathcal{L}(U).$$

Since $\gamma_g/(g + \nu) \leq \Lambda(\nu)$, choose $\varepsilon, \delta > 0$ small enough that

$$(\gamma_g + \delta) \left(\frac{1}{g + \nu} + \varepsilon \right) < A \quad (2 \leq g \leq J).$$

There are only finitely many such comparisons. Rounding changes their double-logarithmic cutoffs by $o(1)$. Each term of (13), including T_k , has rate strictly below A . A fixed finite sum has rate at most the greatest term rate. Letting A decrease to $\Lambda(\nu)$ proves the matching bound.

The input (11) implies $\gamma_{600} = 1$. A given consecutive pair belongs to at most 601 integer-start windows of width 600, so the associated cumulative gap count has a positive-power lower bound along large cutoffs. The trivial bound $G_H(X) \leq X$ gives the reverse inequality on its exponent. Hence $\Lambda(\nu) \geq 1/(600 + \nu) > 0$. Together with the fixed-rank proof this completes Theorem 3.2.

For the canonical upper bound use the minimum eventual gap, two, and $\nu = 1$ in (12). Reducing ε slightly absorbs the rounding, leaving all ascents below $\mathcal{E}((1/3 + \varepsilon)k)$. Their number bounds $N_\eta(k)$. Together with Corollary 5.2, this proves Theorem 3.3.

7 A finite phase profile

Theorem 7.1. *Let H_* be the least $H \geq 2$ for which $\gamma_H = 1$. This integer exists, and for every fixed $\nu > 0$,*

$$\Lambda(\nu) = \max_{2 \leq H \leq H_*} \frac{\gamma_H}{H + \nu}.$$

For all sufficiently large ν ,

$$\Lambda(\nu) = \frac{1}{H_* + \nu}, \quad \frac{1}{\Lambda(\nu)} - \nu = H_*. \quad (14)$$

Proof. Existence follows from $\gamma_{600} = 1$. If $H > H_*$, then $\gamma_H/(H + \nu) \leq 1/(H + \nu) < 1/(H_* + \nu)$. For each $H < H_*$ put $\gamma = \gamma_H < 1$. With positive denominators, the inequality $\gamma/(H + \nu) < 1/(H_* + \nu)$ is equivalent to $(1 - \gamma)\nu > \gamma H_* - H$. It holds beyond a finite threshold. Take the greatest threshold over the finitely many smaller classes. $\square$

Here full frequency means only $\gamma_H = 1$. It need not imply positive relative density or a fixed power lower bound at every large cutoff. The theorem identifies a finite maximum without determining its entries numerically. The rate is an upper envelope of the gap-frequency data; the result recovers H_* , not every individual value of γ_H .

8 Arithmetic realization by affine prime sums

We prove Theorem 3.4 without discarding exceptional primes.

8.1 Fixed-modulus limits and independence

For fixed Q , the prime number theorem in arithmetic progressions gives

$$\frac{\#\{p \leq X : p \text{ prime, } p \equiv a \pmod{Q}\}}{\pi(X)} \longrightarrow \frac{1}{\varphi(Q)} \quad ((a, Q) = 1).$$

Nonunit prime classes contain only primes dividing Q , so their normalized contribution vanishes. Full Cartesian sampling of the ordered pairs makes each fixed pair of unit classes have limiting probability $\varphi(Q)^{-2}$.

Take Q to be the product of any fixed finite set of primes. The Chinese remainder theorem identifies the unit-pair space with the product of the local unit-pair spaces. Divisibility of F at a prime depends only on the local pair, so the limiting divisibility indicators are independent, with exact law

$$\eta_\ell = \frac{\#\{(u, v) \in (\mathbb{F}_\ell^\times)^2 : \prod_j (c_j(u + v) + d_j) = 0\}}{(\ell - 1)^2}. \quad (15)$$

For fixed k, ℓ the rank event uses only the primes at most ℓ . Summing their joint atoms gives (1).

If $F(p, q) = 0$, then some $p + q = -d_j/c_j \leq |d_j| \leq \sum_j |d_j|$. Only finitely many such pairs exist, independently of X . Their normalized contribution is zero. The diagonal was included throughout; its $\pi(X)$ pairs are also negligible against $\pi(X)^2$. No modulus or rank was allowed to grow with X .

8.2 Exact local counts

Set $T_\ell = \{t \in \mathbb{F}_\ell : \prod_j (c_j t + d_j) = 0\}$. There are $\ell - 1$ unit pairs with sum zero, and $\ell - 2$ with any specified nonzero sum. Thus at every prime,

$$\eta_\ell = \frac{(\ell - 2)|T_\ell| + \mathbf{1}_{0 \in T_\ell}}{(\ell - 1)^2}. \quad (16)$$

If a factor is identically zero modulo ℓ , $T_\ell = \mathbb{F}_\ell$ and $\eta_\ell = 1$. An empty root set gives zero, and coincident roots are counted once. The formula covers all coefficient, shift, determinant, and small-prime exceptions.

Outside the finite set dividing the nonzero coefficients, nonzero shifts, or determinants $c_i d_j - c_j d_i$, there are m distinct roots. At most one shift is zero; let z_0 be one if it is and zero otherwise. Then

$$\eta_\ell = \frac{m(\ell - 2) + z_0}{(\ell - 1)^2}, \quad \eta_\ell^{-1} = \frac{\ell}{m} - \frac{z_0}{m^2} + O_F(\ell^{-1}). \quad (17)$$

The law satisfies (4) with $\nu = m$. Theorem 3.2 proves the arithmetic theorem.

The simplest instances illustrate the exceptional coordinate:

$F(p, q)$	η_2	$\eta_\ell \ (\ell > 2)$
$p + q$	1	$1/(\ell - 1)$
$p + q + 1$	0	$(\ell - 2)/(\ell - 1)^2$
$2(p + q) - 1$	0	$(\ell - 2)/(\ell - 1)^2$.

The forced factor 2 in $p + q$ shifts subsequent ranks by one. Both odd examples have exactly the canonical odd law. All three have rate $\Lambda(1)$, despite different finite-rank behavior. Equations (5) and (14) also show how sufficiently large fixed numbers of distinct affine factors recover H_* . There is no uniformity claim when the family varies with the rank. The arithmetic realization uses fixed-modulus prime progression limits. It does not transfer the ordinary-law RH criterion of Section 11 automatically to these families; that extension would require a separate coefficient theorem.

9 Late ascents and bounded prime gaps

For a canonical law, a *late ascent at coefficient b* is an ascent across consecutive primes $p < q$ with $\mathcal{L}(p - 1) \geq bk$. Unbounded supply requires existence beyond every prescribed lower bound on both k and p .

Theorem 9.1 (Recurrence equivalence). *Let $H \geq 1$ be an integer and $1/(H+2) < b < 1/(H+1)$. Gaps at most H recur arbitrarily far out if and only if for every K, Q there exist $k \geq K$ and consecutive primes $p < q$, with $p \geq Q$ and $\mathcal{L}(p-1) \geq bk$, at which both canonical densities ascend.*

Proof. Choose $b < a < 1/(H+1)$. To a sufficiently late gap at most H assign $k = \lfloor \mathcal{L}(p-1)/a \rfloor$. Then $k \rightarrow \infty$ with p , $\mathcal{L}(p-1) \geq ak \geq bk$, and $\mathcal{L}(p-1)/k \rightarrow a$. A small fixed band around a in Lemma 4.2 gives both ascents.

Conversely, $b > 1/(H+2)$ and the one-sided ratio bound exclude gaps at least $H+1$ from sufficiently late ascents. In the odd law the correction to the ordinary threshold has absolute value at most $1/(p-2)$, absorbed by the fixed strict margin. Unbounded lower primes therefore give recurring gaps at most H . $\square$

Corollary 9.2 (Critical location). *Suppose gaps at most $h \geq 1$ recur arbitrarily far out and every sufficiently late consecutive gap is at least h . The critical canonical late-ascent coefficient is $1/(h+1)$: every smaller positive coefficient has unbounded simultaneous ascent supply; every larger coefficient forces both densities to descend for all sufficiently large ranks and lower primes in that region. No assertion at equality is made.*

Proof. Use the forward rank assignment on gaps at most h and the one-sided upper ratio bound on gaps at least h . $\square$

For every fixed $b > 1/5$, an ascent in either canonical law at sufficiently large k, p with $\mathcal{L}(p-1) \geq bk$ must cross a gap strictly below four. Gaps between odd primes are positive even integers, so it must be two. An independently proved unbounded supply would therefore imply infinitely many twin primes. That supply is not established here. A large finite ascent certifies a finite prime pair, not recurrence.

9.1 Infinitude, frequency and the last ascent

For either canonical family, let P_k denote the prime starting its last ascent, which exists for all sufficiently large k . If h is the least recurring prime gap, Corollary 9.2 and the upper cutoff imply

$$\limsup_{k \rightarrow \infty} \frac{\mathcal{L}(P_k)}{k} = \frac{1}{h+1}.$$

Indeed every smaller coefficient occurs at unbounded ranks and endpoints, whereas every larger coefficient is eventually excluded. Bounded-gap infinitude at width H therefore gives the lower location coefficient $1/(H+1)$. For example, taking width 186 as an input gives a coefficient at least $1/187$; it does not identify the least recurring gap. Twin-prime infinitude would give the location limsup exactly $1/3$. These implications concern location. A frequency hypothesis is needed for the corresponding reversal count.

Theorem 9.3 (Conditional last-ascent location). *For either canonical law, let X_k be the largest supported prime p at which a hypothetical twin step satisfies the strict ascent criterion. Suppose that for some $C > 0$,*

$$\pi_2(X) \sim CX/(\log X)^2, \quad \pi_2(X) = \#\{p \leq X : p, p+2 \text{ are prime}\}.$$

Then, for every sufficiently large k , P_k is exactly the last lower twin member at or below X_k , and

$$\frac{P_k}{X_k} \longrightarrow 1, \quad \frac{\mathcal{L}(P_k)}{k} \longrightarrow \frac{1}{3}.$$

The same hypothesis and the count transfer give $\log \log(3 + N_\eta(k))/k \rightarrow 1/3$ for the canonical reversal counts.

Proof. At a supported prefix, adding a positive weight decreases e_{r-1}/e_r by the elementary symmetric log-concavity inequality. The ordinary hypothetical twin threshold is 3; the odd threshold is $3 - 2/(p(p-2))$, which increases with $p > 2$. Thus the supported eligible primes form an initial segment. The uniform ratio estimates and Bertrand's theorem show that its last prime X_k exists for large k and $\mathcal{L}(X_k)/k \rightarrow 1/3$.

For every fixed $\varepsilon \in (0, 1)$, subtracting the two assumed count asymptotics gives

$$\pi_2(X) - \pi_2((1 - \varepsilon)X) \sim C\varepsilon X/(\log X)^2 > 0.$$

First use dyadic coverage in a fixed scale band strictly between $1/5$ and $1/3$. It supplies a twin ascent whose supported prefix serves as an anchor. Every later ascent must be a twin by the $1/5$ exclusion, and lies at most at X_k . Conversely the last twin at or below X_k follows the anchor, has positive support and satisfies the strict criterion. This proves the exact identification. Now the displayed proportional coverage gives $(1 - \varepsilon)X_k < P_k \leq X_k$ eventually, for each fixed ε , and hence both location limits.

The global twin asymptotic also supplies a positive-power number of width-two windows in every sufficiently large dyadic interval. Corollary 5.2 for every $a < 1/3$, together with the canonical upper bound, proves the count limit. $\square$

The usual Hardy–Littlewood twin conjecture supplies a particular positive constant C ; the proof uses no value of that constant. The fixed-tuple asymptotic and the stronger uniform tuple assumptions used for short-interval distribution should be distinguished [12, (1) and Theorem 1]. Bare twin-prime infinitude does not supply the proportional coverage used here. Nor does the displayed asymptotic give a polylogarithmic or square-root bound for $X_k - P_k$ along these particular cutoffs. A limit on $\mathcal{L}(P_k)/k$ also does not imply $P_k/\mathcal{E}(k/3) \rightarrow 1$: a constant shift in the inner exponent already changes the original endpoint by a large multiplicative factor.

10 Prime-product errors and the full coefficient reference

The remaining sections concern the ordinary integer law. They keep the arithmetic cutoff x and the rank r distinct: x tends to infinity, while r is comparable to $\log \log x$. All prime products in these sections include $p \leq x$. For the prefix before an ascent prime p , the corresponding cutoff is $x = p - 1$.

10.1 Classical analytic quantities

Write

$$\theta(x) = \sum_{p \leq x} \log p, \quad \psi(x) = \sum_{p^a \leq x} \log p, \quad \kappa(t) = \frac{1/\log t + 1/(\log t)^2}{t^2}.$$

For sufficiently large x , set

$$\begin{aligned} J(x) &= \int_x^\infty (\psi(t) - t) \kappa(t) dt, \\ K(x) &= \int_x^\infty (\theta(t) - t) \kappa(t) dt, \\ P(x) &= J(x) - K(x) = \int_x^\infty (\psi(t) - \theta(t)) \kappa(t) dt, \\ f(x) &= e^\gamma \log \theta(x) \prod_{p \leq x} (1 - 1/p), \quad F(x) = \log f(x). \end{aligned} \tag{18}$$

The quantitative prime number theorem gives convergence of these integrals. These are Nicolas’s quantities [7, (1.9), (1.15)–(1.19)]; P retains every prime power. Let

$$A(x) = \sum_{p > x} (-\log(1 - 1/p) - 1/p),$$

and, when $\theta(x) > 1$, let

$$D(x) = \frac{\theta(x) - x}{x \log x} - (\log \log \theta(x) - \log \log x).$$

Partial summation and the convergent logarithmic prime-power series give the exact identity

$$F(x) = K(x) - D(x) + A(x) = J(x) - P(x) - D(x) + A(x). \quad (19)$$

Here $A(x) \geq 0$, $A(x) = O(1/x)$, and $D(x) \geq 0$ by concavity of $\log \log t$ on $t > 1$. No sign is assigned to $J(x)$.

We use the following classical facts, with their hypotheses retained. Nicolas's criterion is equivalent to eventual $F(x) < 0$. Under RH,

$$F(x) = -\frac{2 + W(x)}{\sqrt{x} \log x} + O\left(\frac{1}{\sqrt{x}(\log x)^2}\right), \quad |W(x)| \leq \beta_0, \quad (20)$$

where

$$W(x) = \operatorname{Re} \sum_{\rho} \frac{x^{i \operatorname{Im} \rho}}{\rho(1 - \rho)}, \quad \beta_0 = 2 + \gamma - \log(4\pi) < 2.$$

The zeros in this absolutely convergent series are counted with multiplicity. Defining the real projection explicitly matches the maintained formal statement. The expansion and its constant are classical [7, Proposition 2.1]. In the decomposition (19), the zero wave comes from J , while

$$P(x) \sim \frac{2}{\sqrt{x} \log x}$$

holds unconditionally and gives the leading prime-square contribution. Under RH the standard bounds $|\psi(x) - x| + |\theta(x) - x| = O(\sqrt{x}(\log x)^2)$ imply

$$D(x) = O((\log x)^3/x), \quad |F(x) - K(x)| = O((\log x)^3/x). \quad (21)$$

Indeed the second derivative of $\log \log t$ is $-(\log t + 1)/(t^2(\log t)^2)$; Taylor's integral remainder, with $\theta(x)/x \rightarrow 1$, proves the first estimate, and (19) proves the second.

If RH fails, F has excursions of both signs of order x^{-b} for some $0 < b < 1/2$ [6, 7]. Section 11.1 records the zero-dependent version used in the coefficient transfer. In particular, the estimate for D in (21) is never used under the assumption that RH is false.

10.2 An entire profile and two different clocks

Define

$$\begin{aligned} G_x(z) &= \prod_{p \leq x} \left(1 + \frac{z}{p-1}\right), & a_r(x) &= [z^r]G_x(z), \\ \mathcal{H}(z) &= \prod_p \left(1 + \frac{z}{p-1}\right) (1-1/p)^z, & C_r(u) &= [z^r]\mathcal{H}(z)e^{uz}. \end{aligned} \quad (22)$$

The power $(1-1/p)^z$ uses the real logarithm of its positive base. The product for $\mathcal{H}$ converges uniformly on compact subsets of $\mathbb{C}$: the logarithmic first-order terms cancel and the remaining tails are bounded by a constant times $\sum_p p^{-2}$. It defines an entire function, positive on the nonnegative real axis, with $\mathcal{H}(0) = \mathcal{H}(1) = 1$.

The finite generating product is classical in this problem [2, (1.6)]; the normalized Euler factor also appears in the mod-Poisson description of prime divisibility [8, §4]. Our reference retains all its Taylor coefficients. It is not a finite-order truncation of a saddle-point expansion.

Let $\mathcal{H}_x$ be the finite product in (22), and put

$$B(x) = \log G_x(1), \quad L(x) = \gamma + \log \log \theta(x). \quad (23)$$

Then $F(x) = L(x) - B(x)$ and, exactly,

$$G_x(z) = \mathcal{H}_x(z)e^{B(x)z}. \quad (24)$$

For positive supported degrees define

$$R_r(x) = \frac{a_{r-1}(x)}{a_r(x)}, \quad \mathcal{R}_r(u) = \frac{C_{r-1}(u)}{C_r(u)}, \quad \Delta_r(x) = R_r(x) - \mathcal{R}_r(L(x)).$$

The denominator conditions below ensure that these ratios have their ordinary real-valued meaning.

Theorem 10.1 (Uniform coefficient transfer). *Fix $0 < a \leq b < \infty$. There exist X and $K > 0$ such that, for every real $x \geq X$ and integer r with $a \leq r/L(x) \leq b$, both denominators are positive and*

$$\left| \Delta_r(x) - \frac{r}{L(x)^2} F(x) \right| \leq K \left(\frac{|F(x)|}{L(x)^2} + \frac{1}{rx} \right). \quad (25)$$

The constants may depend on a, b , but not on r or x .

Proof. Write $\mathcal{H}(z) = \sum_{j \geq 0} h_j z^j$ and

$$Q_r(u) = \frac{r!C_r(u)}{u^r} = \sum_{j=0}^r h_j \frac{(r)_j}{u^j}.$$

For $t = r/u$ in a fixed positive compact interval, the falling-factorial estimate

$$\left| \frac{(r)_j}{r^j} - 1 + \frac{j(j-1)}{2r} \right| \leq \frac{j^4}{r^2}$$

holds also for $j > r$ if $(r)_j$ is extended by zero. For $j \leq r$, apply the first two inclusion-exclusion bounds to $\prod_{i=0}^{j-1} (1 - i/r)$; the other case is immediate. Cauchy coefficient bounds on a circle of radius larger than the compact interval justify summation with the factor j^4 . Hence

$$Q_r(u) = \mathcal{H}(t) - \frac{t^2 \mathcal{H}''(t)}{2r} + O(r^{-2}). \quad (26)$$

Since $\mathcal{H}(t) > 0$, the normalized denominators are bounded away from zero. Using $C'_r = C_{r-1}$ and the same expansion for $r-1, r-2$ gives

$$\mathcal{R}'_r(u) = \frac{C_{r-2}(u)}{C_r(u)} - \mathcal{R}_r(u)^2 = -\frac{r}{u^2} + O(r/u^3). \quad (27)$$

The first-order profile terms cancel in this paired difference.

On any fixed larger circle, $\mathcal{H}_x - \mathcal{H} = O(1/x)$ by the quadratic prime tail. To compare ratios without losing the cancellation, use

$$\frac{C_{r-1}(u)}{C_r(u)} = \frac{r}{u} \left(1 + \frac{Q_{r-1}(u) - Q_r(u)}{Q_r(u)} \right).$$

The identity $(r)_j - (r-1)_j = j(r-1)_{j-1}$ supplies a factor $1/u$ in the numerator difference. Replacing $\mathcal{H}$ by $\mathcal{H}_x$ therefore changes this ratio by $O(1/(rx))$, uniformly in the compact rank range. Equation (24) yields

$$R_r(x) = \mathcal{R}_r(B(x)) + O(1/(rx)).$$

Unconditionally $F(x) = o(1)$ by PNT and Mertens's theorem. Integrate (27) between L and $B = L - F$. The main term is rF/L^2 ; its error is $O(|F|/L^2)$ since r is comparable to L and $F = o(1)$. This proves (25). $\square$

The coefficient expansion belongs to the saddle-point and mod-Poisson tradition [2, 8, 9]. The sign-preserving comparison (25), its theta centering, and its application to the actual ascent ratio are the potentially new statements here. Any fixed error $O(L^{-m})$ would be larger than the RH signal $1/(L\sqrt{x} \log x)$.

11 RH criteria and quantitative excursions

Theorem 11.1 (Moving-rank RH criterion). *Fix $\alpha > 0$ and set $r(x) = \lfloor \alpha L(x) \rfloor$. Then RH is equivalent to*

$$R_{r(x)}(x) < \mathcal{R}_{r(x)}(L(x)) \quad \text{for every sufficiently large } x.$$

Under RH, more precisely,

$$\frac{L(x)^2}{r(x)} \sqrt{x} \log x \Delta_{r(x)}(x) = -2 - W(x) + o(1). \quad (28)$$

If RH is false, there exists one $b \in (0, 1/2)$ such that, for every fixed $\alpha > 0$,

$$\limsup_{x \rightarrow \infty} x^b L(x) \Delta_{r(x)}(x) = +\infty, \quad \liminf_{x \rightarrow \infty} x^b L(x) \Delta_{r(x)}(x) = -\infty. \quad (29)$$

The witnessing endpoints may depend on α and on the sign.

Proof. The floor rank is eventually in a fixed positive compact band. Under RH, substitute (20) into (25). After multiplication by $(L^2/r)\sqrt{x} \log x$, the two errors are respectively $O(1/L)$ and $O((\log x)/\sqrt{x})$. Both vanish, giving (28). Since $2 - \beta_0 > 0$, the sign is eventually negative.

Under false RH, Theorem 11.2 below supplies b and excursions of F with arbitrarily large amplitudes. In a band $a \leq r/L \leq b_1$, (25) gives, eventually,

$$\begin{aligned} F \geq 0 &\implies L\Delta_r \geq (a/2)F - K/(ax), \\ F \leq 0 &\implies L\Delta_r \leq (a/2)F + K/(ax). \end{aligned}$$

Here b_1 denotes the fixed band endpoint, not the excursion exponent. For the floor rank one may take $a = \alpha/2$. Since $x^{-1} = o(x^{-b})$, both excursions transfer with arbitrarily large amplitudes. This proves (29) and the reverse implication. $\square$

The equality (28) is an ordinary corollary of the compiled linearization and spectral estimates. The maintained criterion and the quantitative false-RH consumer are separate named declarations; the ledger distinguishes these

verification scopes.

11.1 The corrected logarithm at power-profile peaks

Theorem 11.2 (Zero-dependent excursions). *Suppose $\zeta(\rho) = 0$ with $1/2 < \operatorname{Re} \rho < 1$. For every*

$$1 - \operatorname{Re} \rho < b \leq 1/2$$

and each $U \in \{J, K, F\}$,

$$\limsup_{x \rightarrow \infty} x^b U(x) = +\infty, \quad \liminf_{x \rightarrow \infty} x^b U(x) = -\infty.$$

Consequently, false RH supplies a common exponent $0 < b < 1/2$.

Proof. We give the analytic mechanism and the endpoint correction explicitly. For $a = \log 3$, initially in $\operatorname{Re} s > 0$, put

$$\widehat{J}(s) = \int_a^\infty J(e^u) e^{-su} du, \quad T(z) = \int_3^\infty (\psi(t) - t) t^{-z-1} dt.$$

The Mellin identity is $T(z) = -\zeta'(z)/(z\zeta(z)) - 1/(z-1)$ minus the entire contribution of the finite interval from 1 to 3. Integration by parts gives

$$s\widehat{J}''(s) + 2\widehat{J}'(s) = T'(s+1) - T(s+1) + a^2 J(3) e^{-as}. \quad (30)$$

The cancellation at $z = 1$ and the quantitative PNT extend $\widehat{J}$ across $s = 0$ and along the real interval $s > -1/2$. At $s = \rho - 1$, the double pole from T' cannot cancel with the simple pole from T . Thus $\widehat{J}$ cannot be holomorphic there.

An eventual bound $J(x) \geq -Cx^{-b}$ would make $J(e^u) + Ce^{-bu}$ eventually nonnegative. Its Laplace transform has the preceding nonreal singularity, but is analytic on the real interval $s > -b$. Landau's singularity principle contradicts this: its abscissa of convergence lies between $\operatorname{Re} \rho - 1$ and 0, hence strictly above $-b$. Applying the same argument to $-J$ rules out the opposite bound. This is the classical Landau method; see [10, Theorem 3]. Since $P(x) = O(x^{-1/2}/\log x) = o(x^{-b})$, $K = J - P$ inherits both signs.

For negative excursions, $F \leq K + A$ and $A = O(1/x)$ suffice. For positive excursions one must control D without RH. Fix $A_0 > 0$, and choose an integer local maximum $x = n + 1$, $n \geq 2$, of the sequence $K(n) - A_0 n^{-b}$ between two lower values bracketing a high positive value. Such maxima exist arbitrarily far out because of the two unbounded excursions of K ; rounding real endpoints to integers costs $O((\log x)/x) = o(x^{-b})$.

The two adjacent comparisons are

$$K(n) - A_0 n^{-b} \leq K(n+1) - A_0 (n+1)^{-b},$$

$$K(n+2) - A_0 (n+2)^{-b} \leq K(n+1) - A_0 (n+1)^{-b}.$$

On each open unit interval θ is constant. Using $K(v) - K(w) = \int_v^w (\theta(t) - t)\kappa(t) dt$ and the possible endpoint jump of at most $\log x$ gives

$$0 < \theta(x) - x \leq (2A_0 + 1)x^{1-b} \log x.$$

Taylor's integral remainder for $\log \log$ and $A(x) \geq 0$ now yield

$$F(x) \geq K(x) - \frac{(2A_0 + 1)^2}{2} (\log x + 1)x^{-2b}.$$

The loss is $o(x^{-b})$. Choosing A_0 larger than any prescribed amplitude proves the positive excursions of F . Finally, if RH fails, symmetry of the zeros supplies such a ρ ; choose any b strictly between $1 - \operatorname{Re} \rho$ and $1/2$. $\square$

The existence of false-RH oscillations of F is classical [6, 7]. The displayed proof records the zero-dependent exponent and the exact nonlinear correction used by the formal coefficient consumer.

11.2 Integral, clock and positive-tilt criteria

Define

$$\mathcal{C}(x) = \exp(e^{-\gamma} G_x(1)).$$

The exact identity $\mathcal{C}(x) = \theta(x)^{1/f(x)}$ shows that eventual $\theta(x) < \mathcal{C}(x)$ is another form of Nicolas's criterion. Combining the preceding results gives

$$\text{RH} \iff K(x) < 0 \text{ eventually} \iff \theta(x) < \mathcal{C}(x) \text{ eventually.} \quad (31)$$

Under RH,

$$\frac{\mathcal{C}(x) - \theta(x)}{\sqrt{x}} = 2 + W(x) + o(1). \quad (32)$$

Indeed $\log \mathcal{C} - \log \theta = (1/f - 1) \log \theta$; (20), $\theta(x)/x \rightarrow 1$, and $e^t = 1 + t + O(t^2)$ prove (32). This is a change of variables in the classical Nicolas estimate.

For fixed $z > 0$, put

$$T_z(x) = \log \left(\frac{e^{\gamma z} \mathcal{H}(z) (\log \theta(x))^z}{G_x(z)} \right).$$

Then, exactly,

$$T_z(x) = zF(x) + \sum_{p>x} \left\{ \log \left(1 + \frac{z}{p-1} \right) - z \log \left(1 + \frac{1}{p-1} \right) \right\}. \quad (33)$$

For fixed z the tail is $O_z(1/x)$. It is nonpositive for $z \geq 1$, and lies between 0 and Cz/x for $0 < z \leq 1$, with an absolute C . Concavity of $\log(1+zw)$ proves the signs; the upper bound follows from $\log(1+zw) \leq zw$ and $w - \log(1+w) \leq w^2/2$. Therefore

$$\text{RH} \iff T_z(x) < 0 \text{ eventually}$$

for each fixed $z > 0$. Under RH a *single* sufficiently large cutoff works simultaneously for all $z > 0$, by splitting at $z = 1$ in (33); the converse follows from $T_1 = F$.

The corresponding positive kernel is

$$\begin{aligned} \kappa_z(t) &= \frac{z}{(t-1)(t-1+z) \log t} + \frac{\log(1+z/(t-1))}{t(\log t)^2}, \\ I_z(x) &= \int_x^\infty (\theta(t) - t) \kappa_z(t) dt. \end{aligned}$$

For $t \geq e$, $|\kappa_z(t) - z\kappa(t)| \leq 14(z+z^2)/(t^3 \log t)$. Chebyshev's bound gives $I_z = zK + O_z(1/(x \log x))$. It follows that eventual $I_z < 0$ is equivalent to RH and that

$$\sqrt{x} \log x I_z(x) = -z(2 + W(x)) + o(1)$$

under RH. Under false RH every fixed positive tilt inherits the zero-dependent two-sided integral excursions. These tilt criteria transport the same leading error; they do not supply independent sign information by themselves.

12 Reference crossings and RH ascent capacities

The preceding centered criterion concerns $R_r - \mathcal{R}_r(L)$. A raw ascent instead concerns $R_r - (g+1)$ for an actual gap g . We now keep both quantities in the comparison.

Theorem 12.1 (Sharp reference crossing). *For every fixed $s > 0$ there are $A, C > 0$ and r_0 such that, for $r \geq r_0$, the equation $\mathcal{R}_r(u) = s$ has a unique solution $u_{r,s}$ in $[r/s - A, r/s + A]$, and*

$$u_{r,s} = \frac{r}{s} - \frac{\mathcal{H}'(s)}{\mathcal{H}(s)} + O_s(1/r). \quad (34)$$

Proof. Expanding the adjacent normalized coefficients as in (26) gives, for $t = r/u$ in a compact neighborhood of s ,

$$\mathcal{R}_r(u) = t - \frac{t^2}{r} \frac{\mathcal{H}'(t)}{\mathcal{H}(t)} + O(r^{-2}).$$

Its values on either side of r/s at a sufficiently large fixed distance straddle s . The derivative (27) is strictly negative throughout that interval, giving existence and uniqueness. Substitute $u = r/s + v$, with bounded v , into the last display to obtain (34). $\square$

For $s = h + 1$, put $X_{r,h} = \mathcal{E}(u_{r,h+1} - \gamma)$. The reference crossing admits a local spatial interpretation. Fix a comparison constant $D_0 \geq 1$. In the window

$$B(x) \in [r/(h+1) - A, r/(h+1) + A], \quad \mathcal{C}(x) \leq D_0 x, \quad X_{r,h} \leq D_0 x,$$

the maintained coefficient comparison gives, for sufficiently large r, x and a constant K depending on the fixed parameters,

$$\begin{aligned} X_{r,h} - \mathcal{C}(x) > K \log x &\implies R_r(x) > h + 1, \\ \mathcal{C}(x) - X_{r,h} > K \log x &\implies R_r(x) < h + 1. \end{aligned} \tag{35}$$

Indeed $R_r(x) - \mathcal{R}_r(B(x)) = O(1/(rx))$, whereas $|\mathcal{R}'_r| \gg 1/r$ in the window. The uncertainty in B is $O(1/x)$; the derivative of $\mathcal{E}(u - \gamma)$ is $\mathcal{E}(u - \gamma) \log \mathcal{E}(u - \gamma)$, giving the stated spatial margin. For an actual gap h , these are ascent/descent tests. Equation (32) locates their RH transition in the θ coordinate, with displacement $(2 + W(x))\sqrt{x} + o(\sqrt{x})$.

Theorem 12.2 (RH endpoint and reversal-count envelope). *Assume RH. Let P_k be the lower prime of the last ordinary ascent and $N_k = N_{\text{ordinary}}(k)$. For all sufficiently large k , both are attained, and with $r = k - 1$,*

$$\theta(P_k - 1) < \mathcal{E}(u_{r,3} - \gamma). \tag{36}$$

For every $\varepsilon > 0$, eventually

$$(1 - \varepsilon)N_k \log N_k \leq \theta(P_k - 1), \quad N_k \leq (1 + \varepsilon) \frac{\mathcal{E}(u_{r,3} - \gamma)}{\log \mathcal{E}(u_{r,3} - \gamma)}.$$

There is a constant C such that

$$\log \log N_k \leq \frac{k-1}{3} - \gamma - \frac{\mathcal{H}'(3)}{\mathcal{H}(3)} + \frac{C}{k-1} \tag{37}$$

for all sufficiently large k .

Proof. The unconditional lower reversal bound supplies a last ascent whose endpoint is at least double-exponential in a positive multiple of k . The unconditional upper cutoff puts it below $\mathcal{E}((1/3 + \varepsilon)k)$ for every fixed $\varepsilon > 0$. PNT therefore places $r/L(P_k - 1)$ in a fixed positive compact band. Under RH, Theorem 10.1 and (20) give $R_r(P_k - 1) < \mathcal{R}_r(L(P_k - 1))$ uniformly in that band. The actual ascent requires $R_r > g + 1 \geq 3$.

Near $u_{r,3}$, the reference is strictly decreasing. Away from its fixed neighborhood the uniform leading expansion separates its value from 3. These two comparisons imply $L(P_k - 1) < u_{r,3}$, which is (36). Every separated witness uses a distinct ascent index, so $N_k \leq \pi(P_k)$. PNT yields the first two count inequalities. Equivalently one can count all primes with $\theta(p-1)$ below the reference endpoint; this monotone capacity has asymptotic $T/\log T$ as $T \rightarrow \infty$.

Taking two logarithms of that capacity gives $u_{r,3} - \gamma + o(1/r)$: the correction from division by $\log T$ is negligible on this scale, and the multiplicative PNT error is bounded eventually. Equation (34) now proves (37). $\square$

The same comparison keeps the actual gap of each ascent: in a fixed positive compact rank band, RH implies

$$g_i + 1 < \mathcal{R}_{k-1}(L(p_i - 1)).$$

Thus, on any sufficiently late finite band of prime indices, the number of ascents is at most the number of indices satisfying this reference inequality with their *actual* g_i . Adding the number of earlier primes bounds every reversal family. Combining this upper count with the unconditional reversal supply forces many reference-eligible actual gaps, but does not lower their maximum width without an additional distribution argument.

Neither the square-root clock displacement nor the $O(1/k)$ count correction changes a leading coefficient measured by $\log \log X/k$. The rank is fixed in the raw reversal count; it moves with x in Theorem 11.1.

13 Implications and scope

The spectrum gives a quantitative dictionary. Bounded-gap counts yield reversal counts through the finite selection lemma; the upper comparison shows that no other late ascents can improve the rate. Finite changes in an admissible law preserve that rate. Many different affine prime-pair expressions consequently share a spectrum despite different forced primes and small ranks.

Location and frequency encode different information. Corollary 9.2 depends on the least recurring gap. The rate $\Lambda(\nu)$ depends on how frequently bounded gap classes occur. Infinitude alone need not give $\gamma_H > 0$: a hypothetical count growing as a power of $\log X$ has $\gamma_H = 0$. This is why bare infinitude cannot replace the input to Corollary 5.2.

For Goldbach-type expressions, the arithmetic theorem describes factorization in the full representation-weighted ensemble. It gives no lower bound for representations of each fixed even integer. Fixed-modulus independence likewise does not supply uniform estimates when a prime cutoff grows with input size. Moving-tail distribution laws require additional uniformity.

Repeated raw reversals and centered moving-rank excursions have different quantifiers. At each fixed rank there is a final decreasing tail. The RH criterion lets the rank grow with the endpoint and compares its ratio with the full reference profile. It is the transfer in Theorem 10.1 that relates this second question to Nicolas's error.

The exact raw comparison can be written as

$$R_r(x) - (g + 1) = (\mathcal{R}_r(L(x)) - (g + 1)) + \Delta_r(x).$$

RH controls the second term. The first term and the occurrence of the actual gap remain necessary for an ascent. In particular, for fixed $1/5 < b < c < 1/3$ and all sufficiently large k , an ascent at a prime with $bk \leq \mathcal{L}(p - 1) \leq ck$ occurs exactly when the next gap is two. This follows directly from the uniform ratio margins. Proving positive ascent supply in this window independently would prove twins; the identity itself is not such a supply proof.

The Hardy–Littlewood input in Theorem 9.3 provides actual twins approaching the eligibility cutoff. RH instead refines that cutoff's reference error and gives Theorem 12.2. The global twin asymptotic allows a distance $o(X_k)$ from the cutoff, which can exceed $\sqrt{X_k}$; it cannot by itself transfer the RH square-root clock correction to the location of the last actual ascent. No implication from RH to twin-prime infinitude is established by these results.

The relation to Robin and Lagarias is through their established equivalences with RH [11]. Thus the eventual inequality in Theorem 11.1, if proved independently, would imply the corresponding universal divisor-sum inequalities. A valid violation of either criterion would imply the false-RH coefficient excursions. These are logical consequences of known equivalences, not new bounds for colossally abundant numbers. Indeed, for $n = \prod p^{a_p}$,

$$\frac{\sigma(n)}{n} = \prod_{p|n} \frac{1 - p^{-(a_p+1)}}{1 - p^{-1}},$$

and optimizing the exponent-dependent numerator factors together with $\log \log n$ remains a separate problem.

The classical inputs include Nicolas’s criterion and spectral formula, Landau oscillation, prime-factor generating products and saddle-point estimates. The potentially new contributions are the exact reversal-frequency spectrum, its affine realization and the sign-preserving moving-coefficient transfer with its applications. These attributions are provisional: proof verification does not establish priority. Speculative RH and independent small-gap-supply investigations are excluded from the theorem claims of this paper.

14 Formalization and reproducibility

The principal maintained owners are:

- `Assembly/ReciprocalSmoothSpectrum.lean`: all-rank finite maxima and Theorem 3.2;
- `Assembly/Headline.lean`: Theorem 3.3;
- `Assembly/AffineSpectrum.lean`: Theorem 3.4 and arithmetic phase recovery;
- `Assembly/GapSpectrumPhase.lean`: Theorem 7.1;
- `Assembly/GapRecurrence.lean` and `Proof/Upper/GapScaleCutoff.lean`: Theorem 9.1, Corollary 9.2, and the conditional inverse twin-prime transfer;
- `Helpers/PrimeProfileCanonicalLinearization.lean` and `Helpers/PrimeProfileUniformSign.lean`: the full-profile coefficient comparison and uniform sign transfer;
- `Helpers/NicolasPowerPeak.lean` and `Helpers/PrimeProfilePowerExcursions.lean`: the corrected logarithm and actual moving-rank power excursions;
- `Helpers/NicolasLogSpectral.lean`, `Helpers/NicolasSpatialClock.lean` and the positive-tilt modules: the classical spectral inputs and their clock/integral consumers;
- `Assembly/RHSharpAscentReference.lean` and `Assembly/RHLogAscentCapacity.lean`: the maintained endpoint and count envelopes.

Paths are relative to `PrimeFactorOscillations/`. The theorem ledger maps intermediate arguments to declarations. Ordinary prose sometimes combines several compiled lemmas; these combinations are distinguished from individually named declarations.

The analytic inputs are the reciprocal-prime estimate, fixed-modulus prime progression limits and (11). Narrow leaves of the pinned Erdős 690 development supply ordinary density and symmetric bounds; its conditional classification is not an input. The width-600 count is extracted from a separately pinned prime-cluster development. The RH part additionally reuses the Robin1984 analytic development and the verified explicit-formula and complex-analytic dependencies. Classical analytic results are retained as proved dependencies, not research-target axioms. The paper provides the coefficient-transfer argument and its consumers, while citing the underlying classical prime-distribution and Nicolas theorems.

The relative-location implication in Theorem 9.3 has a compiled conditional research consumer; its maintained public export is still pending. Its full written proof is included above. The resulting logarithmic location and count limits are ordinary consequences of that proof and the maintained transfers. Likewise (28) combines verified estimates, but is not advertised as a separately compiled normalized-wave declaration. These qualifications concern formal integration, not an assumption that Hardy–Littlewood or RH has been proved.

The public Challenge states two headline theorems over Mathlib primitives. Solution contains their actual proofs. The two deliberate Challenge placeholders are absent from the Solution closures. The 4 October proof audit checks the two public compiled statement pairs and 218 selected theorem closures, finding only `propext`, `Classical.choice`, and `Quot.sound`. The RH statements have maintained consumers in Solution with their RH, false-RH or zero hypotheses explicit. An axiom audit does not remove any of these hypotheses. The two Challenge statements concern the original reversal and affine-spectrum headlines; they are not claimed to be a Challenge comparison for every later RH declaration.

The dependency record and build guide specify Windows Lean 4.34.0, source revisions and artifact hashes. The project reuses the Erdős dependency’s published cache without rebuilding or redistributing it. Additional analytic compatibility artifacts still require an existing matching view; a complete fresh-machine bootstrap is not claimed. The verification record separates the initial publication, the completed 4 October analytic proof audit, and the present exposition and PDF checks. Branch links provide mutable navigation; the records retain checked source hashes.

No numerical experiment proves an asymptotic step here. The exact inequality $1/602 < 1/601$ supplies the canonical constant's strict margin; no effective onset rank is given. Official Comparator/NanoDa replay and independent expert review have not been claimed. Kernel checking verifies proof terms for the stated definitions under the reported foundations. Source fidelity, attribution, novelty and significance require scrutiny beyond the kernel. AI assistance was used for research, proof development and exposition.

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